MATH 11 — ELEMENTARY STATISTICS

§8.1

Introduction to Confidence Intervals

A range of plausible values for a population parameter

Today's Goal

After completing this section, you should be able to:

- Distinguish between point estimates and interval estimates
- Understand the concept of confidence level
- Identify the components of a confidence interval

Learning Target 13

LT.13 Confidence Intervals

I can determine an appropriate confidence interval estimate for the proportion, mean, and standard deviation of a population and I can interpret what the confidence interval means in context.

A Park Survey

A city surveys 400 randomly selected residents, and 52% of them say they support a new park.

Do exactly 52% of all residents in the city support the park?

Different samples will produce different point estimates.

Point Estimates

Definition 8.1.1

A **point estimate** is a single value used to estimate a population parameter.

Common point estimates include:

- The sample mean $\bar{x}$ is the best point estimate of the population mean μ.
- The sample proportion $\hat{p}$ is the best point estimate of the population proportion p.
- The sample variance s^2 is the best point estimate of the population variance σ^2.

The problem with point estimates is that they rarely equal the exact value of the population parameter.

Interval Estimates

Instead of using a single value, we can construct an interval that is likely to contain the true population parameter.

Definition 8.1.2

A **confidence interval** (CI) is a range of values used to estimate the true value of a population parameter.

A confidence interval has the general form:

$$\text{point estimate} - E < \text{parameter} < \text{point estimate} + E$$

where E is called the **margin of error**. The margin of error determines how wide the interval is.

Margin of Error

Definition 8.1.3

The **margin of error** is the maximum likely difference between the point estimate and the true population parameter.

We can also write a confidence interval in interval notation as

$$(\text{point estimate} - E,\ \text{point estimate} + E)$$

or as

$$\text{point estimate} \pm E.$$

Writing a Confidence Interval

(a) A sample has $\bar{x} = 72$, and the margin of error is $E = 4$. Write the confidence interval for μ in all three forms.

$68 < \mu < 76$

$(68,\ 76)$

72 ± 4

(b) A confidence interval for p is $(0.41,\ 0.49)$. Find the point estimate and the margin of error.

Point estimate: $\hat{p} = \frac{0.41+0.49}{2} = 0.45$

Margin of error: $E = \frac{0.49-0.41}{2} = 0.04$

Confidence Level

Every confidence interval is associated with a **confidence level**, which indicates how confident we are that the interval contains the true population parameter.

Definition 8.1.4

The **confidence level** is the probability $1 - \alpha$ (such as 0.95, or 95%) that the confidence interval actually contains the population parameter, assuming that the estimation process is repeated a large number of times.

Common confidence levels are 90%, 95%, and 99%.

Confidence Level

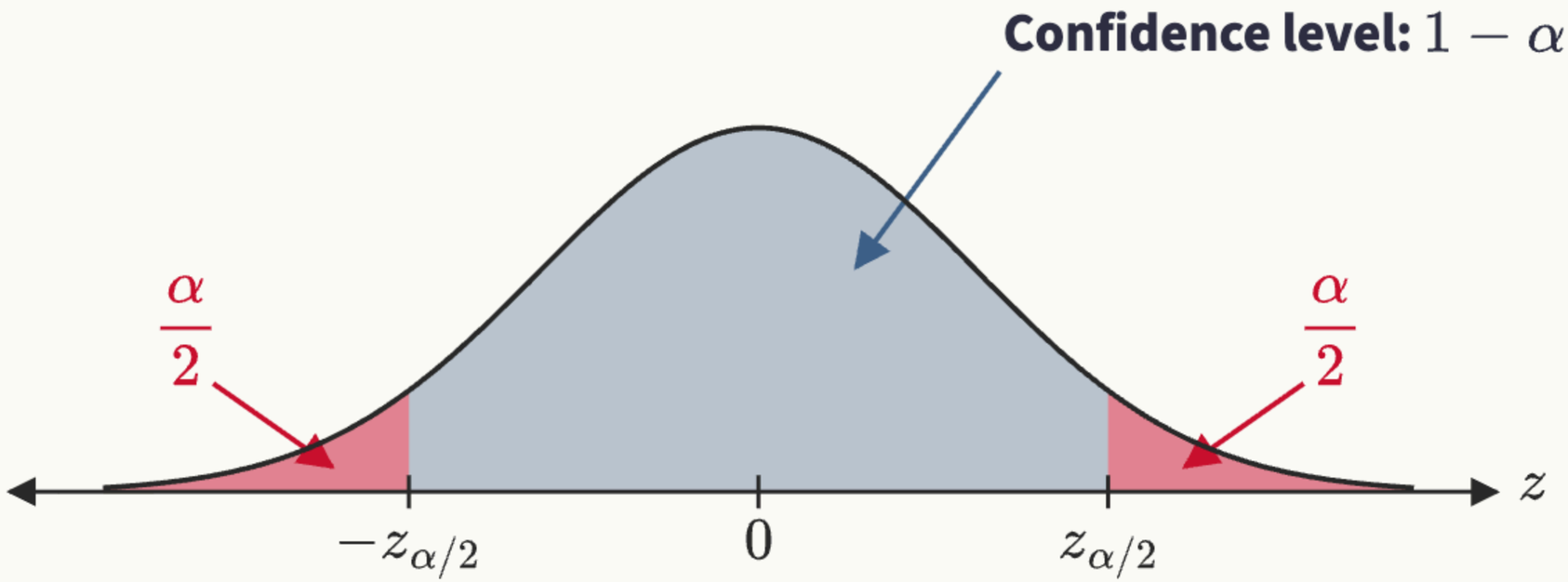


Figure 8.1.5. The confidence level $1 - \alpha$ corresponds to the central area of the distribution; the remaining area α is split equally between the two tails as $\frac{\alpha}{2}$ each.

Interpreting Confidence Intervals

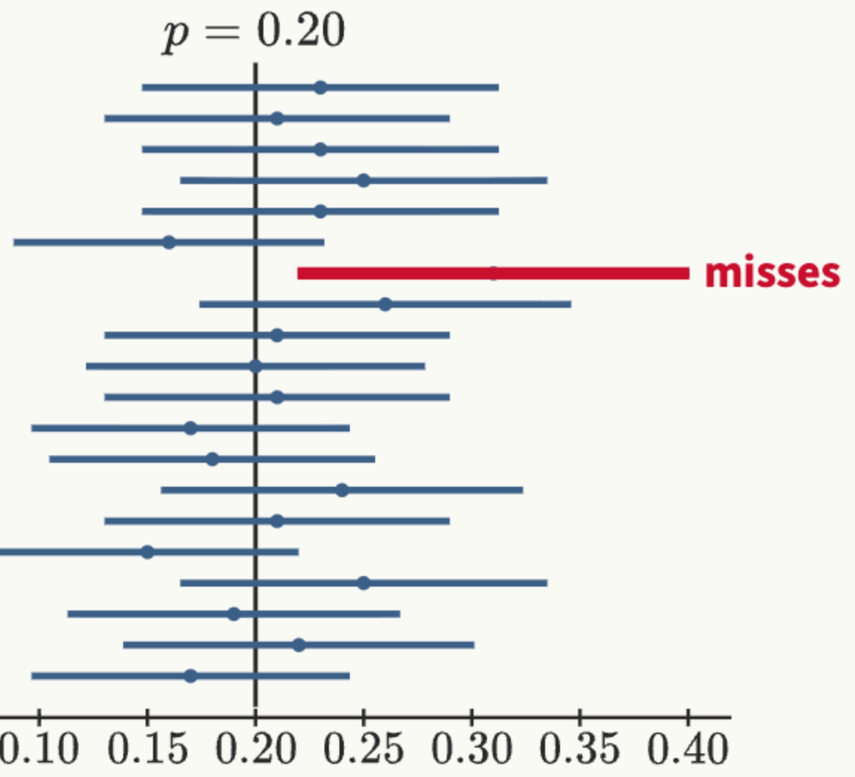
Correct interpretation: If we were to repeat the sampling process many times and construct a confidence interval each time, then 95% of those intervals would contain the true population parameter.

Incorrect interpretation: There is a 95% probability that the true population parameter is within the interval.

The distinction is subtle but important. Once a specific confidence interval is calculated, the population parameter either is or is not within that interval—there is no probability involved. The confidence level refers to the long-run success rate of the method, not to any particular interval.

Twenty Confidence Intervals

A computer simulated 20 random samples of 100 people from a population with $p = 0.20$. Each segment is the 95% confidence interval from one sample. Its dot is $\hat{p}$.



How many of these intervals contain p?

19 of the 20. Each interval either contains p or does not. The 95% describes the method over many samples.

Critical Values

To construct a confidence interval, we need to find the appropriate **critical value**, which depends on the confidence level.

Definition 8.1.6

For a confidence level of $1 - \alpha$, the corresponding **critical value** separates the middle $1 - \alpha$ area from the outer α area of the distribution. We denote this critical value as $z_{\alpha/2}$ or $t_{\alpha/2}$, depending on which distribution we use.

Critical Values

The following table shows critical values (rounded to four decimal places) for common confidence levels:

Table 8.1.7. Common confidence-level critical values.

Confidence Level	α	$z_{\alpha/2}$
90%	0.10	1.6449
95%	0.05	1.960
99%	0.01	2.5758